

VANIER COLLEGE · ADVANCED NETWORKS

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Lab 4: VLANs, Trunking & Inter-VLAN Routing

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General instructions

- Open the provided Packet Tracer file (.pkt) and build your network directly from it. Do not start from scratch.
- For the written document, values from the previous lab are already filled in below. Only new fields are left blank.
- Name the new devices you add consistently with the current network names in the provided PT file.
- Packet Tracer has no GUI option for router subinterfaces, so Step 3 gives you the commands to type.

Scenario

A VLAN separates traffic at Layer 2. An IP subnet separates networks at Layer 3. In a typical enterprise design, each VLAN is assigned its own IP subnet, so a router or Layer 3 switch can route traffic between VLANs.

The company added a Finance department (10 hosts) and moved the main office (MO) to an open-plan floor. One switch has broken, so only two switches are now available in MO. The router in the main office (R-MO) can spare just one link for all four departments. VLANs let these departments share that hardware and stay logically separate.

Given requirements

Segment	Hosts needed	Notes
Sales	up to 50	—
Engineering	up to 20	Includes Web-Eng, a web server reachable by every department
Management	up to 10	—
Finance (new)	up to 10	PCs split across both switches
Remote	up to 4 clients	DHCP from DHCP-Remote. See the note under this table.
R-MO – R-RO WAN	2	Point-to-point link. No VLAN.

Remote stays 192.168.10.112/29 from the previous lab. That subnet has six usable addresses (.113–.118). One is R-RO's gateway and one is the new DHCP-Remote server, so four addresses remain for clients. The old "up to 5" count no longer fits once the server has its own address.

Step 1: Design the VLAN and addressing plan

A VLAN ID is a label a switch or router uses to keep one department's traffic separate from another's, even when they share the same wires. Use VLAN IDs 10, 20, 30, and 40 for Sales, Engineering, Management, and Finance respectively.

Sales, Engineering, and Management keep their original addressing. Write only the VLAN ID for those three rows.

Finance is new and needs 10 hosts, so it gets a /28 (mask 255.255.255.240), the same size as Management. Remote (192.168.10.112/29) and the WAN (192.168.10.120/30) already occupy 192.168.10.112 through 192.168.10.123. Find the next available space for Finance subnet. A gap is normal with VLSM, in case you encounter one.

VLAN ID	Segment	Network address	Subnet mask	Gateway
10	Sales	192.168.10.0	255.255.255.192	192.168.10.1
20	Engineering	192.168.10.64	255.255.255.224	192.168.10.65
30	Management	192.168.10.96	255.255.255.240	192.168.10.97
40	Finance (new)	192.168.10.128	255.255.255.240	192.168.10.129

Step 2: Configure the switches

An **access** port carries one VLAN, untagged. Use access mode for every end device (PCs and servers). A **trunk** carries several VLANs, and each frame is tagged with its VLAN ID. Trunks are the switch-to-switch link and the switch-to-router link.

Create VLANs 10, 20, 30, and 40 on both SW-A and SW-B (Config > VLAN Database) before you assign ports. Do this even when a switch has no access port in that VLAN. A trunk forwards a VLAN only if the VLAN exists on that switch. SW-A has no Engineering or Management hosts, but those VLANs still cross SW-A to reach R-MO. If VLAN 20 or 30 is missing on SW-A, Sales cannot reach Web-Eng.

SW-A Fa0/23, SW-A Fa0/24, and SW-B Fa0/23 are static trunks and must allow VLANs 10, 20, 30, and 40. Set both ends of the SW-A–SW-B link. Fa0/24, toward R-MO, has to be a static trunk. A router does not speak DTP, so a port left as dynamic auto or dynamic desirable stays an access port and only one VLAN will pass.

Switch	Port(s)	Mode	VLAN ID	Notes
SW-A	Fa0/1 – Fa0/2	Access	10	Access. Sales.
SW-A	Fa0/3	Access	10	Access. Finance. Add Finance-PC1 here.
SW-A	Fa0/23	Trunk	10, 20, 30, 40	Static trunk. VLANs 10, 20, 30, 40. Uplink to SW-B.
SW-A	Fa0/24	Trunk	10, 20, 30, 40	Static trunk, not dynamic. VLANs 10, 20, 30, 40. Uplink to R-MO.
SW-B	Fa0/1 – Fa0/3	Access	20	Access. Engineering, including Web-Eng.
SW-B	Fa0/4 – Fa0/5	Access	30	Access. Management.
SW-B	Fa0/6	Access	40	Access. Finance. Add Finance-PC2 here.

Switch	Port(s)	Mode	VLAN ID	Notes
SW-B	Fa0/23	Trunk	10, 20, 30, 40	Static trunk. VLANs 10, 20, 30, 40. Uplink to SW-A.

Write Access or Trunk in Mode. For an access port, write that department's VLAN ID. For a trunk, write 10, 20, 30, 40.

Step 3: Configure router-on-a-stick (CLI)

A **subinterface** makes one physical link act as several virtual links. Each subinterface handles one VLAN's tagged traffic, which is how a single cable from SW-A carries all four departments. The switch sends VLAN-tagged frames over the trunk. The router uses the VLAN ID on each subinterface to decide which Layer 3 network the traffic belongs to.

On R-MO, open the CLI from the Config tab. Two placeholders in the commands below are not the same thing. The number after the dot is a subinterface number: any unique number on that interface. Using the VLAN ID (.10, .20, .30, .40) is the usual choice. The number after encapsulation dot1Q is the VLAN ID, in this order: 10, 20, 30, 40. Only the encapsulation value has to match the VLAN. Do not type the angle brackets. Replace every placeholder before you press Enter.

Use the interface name the .pkt actually shows. It may be GigabitEthernet or FastEthernet. Before you add the subinterfaces, clear any Lab 3 address that is still on R-MO. If 192.168.10.1, 192.168.10.65, or 192.168.10.97 is still configured on this interface or on any other R-MO interface, IOS will reject the new address as an overlap.

```
enable
configure terminal
interface <interface facing SW-A>
  no ip address
  no shutdown
exit
interface <interface facing SW-A>.<subinterface number>
  encapsulation dot1Q <VLAN ID>
  ip address 192.168.10.1 255.255.255.192
exit
interface <interface facing SW-A>.<subinterface number>
  encapsulation dot1Q <VLAN ID>
  ip address 192.168.10.65 255.255.255.224
exit
interface <interface facing SW-A>.<subinterface number>
  encapsulation dot1Q <VLAN ID>
  ip address 192.168.10.97 255.255.255.240
exit
interface <interface facing SW-A>.<subinterface number>
  encapsulation dot1Q <VLAN ID>
  ip address 192.168.10.129 255.255.255.240
exit
end
```

Step 4: Assign PC and server addresses

Sales, Engineering, and Management PCs keep their original addresses. Add each one's VLAN. Web-Eng, the two Finance PCs, and DHCP-Remote are new. Add Finance-PC1 on SW-A Fa0/3 and Finance-PC2 on SW-B Fa0/6.

Web-Eng is an Engineering host: VLAN 20, mask 255.255.255.224, gateway 192.168.10.65, and any unused address in 192.168.10.64/27. Each Finance PC uses VLAN 40, mask 255.255.255.240, gateway 192.168.10.129, and an unused address in 192.168.10.128/28 other than the gateway. DHCP-Remote is not in a MO VLAN. Give it a static address in 192.168.10.112/29 (mask 255.255.255.248) that is not R-RO's gateway. The server's own default gateway is that same R-RO address. Remote-PC1 and Remote-PC2 stay on DHCP and are not in a MO VLAN either.

Device	VLAN	IP address	Subnet mask	Default gateway
Sales-PC1	10	192.168.10.2	255.255.255.192	192.168.10.1
Sales-PC2	10	192.168.10.3	255.255.255.192	192.168.10.1
Engineering-PC1	20	192.168.10.66	255.255.255.224	192.168.10.65
Engineering-PC2	20	192.168.10.67	255.255.255.224	192.168.10.65
Web-Eng	20	192.168.10.68	255.255.255.224	192.168.10.65
Management-PC1	30	192.168.10.98	255.255.255.240	192.168.10.97
Management-PC2	30	192.168.10.99	255.255.255.240	192.168.10.97
Finance-PC1 (SW-A)	40	192.168.10.130	255.255.255.240	192.168.10.129
Finance-PC2 (SW-B)	40	192.168.10.131	255.255.255.240	192.168.10.129
DHCP-Remote (server)	none	192.168.10.116	255.255.255.248	192.168.10.113
Remote-PC1	none	DHCP	DHCP	DHCP
Remote-PC2	none	DHCP	DHCP	DHCP

Step 5: Add DHCP-Remote

Add a Server-PT to the Remote LAN. Give it the static address you chose in Step 4, then enable Services > DHCP. Include a DNS server in the pool. That setting is what lets DHCP clients resolve names, not just reach the network. This lab is not connected to the internet. You are practicing the setting a real pool would hand out, such as an ISP resolver or a public resolver like 8.8.8.8.

The pool's default gateway is R-RO's existing Remote LAN address from Lab 3. It is not 192.168.10.1 and it is not an R-MO subinterface. The subnet mask is 255.255.255.248. Keep every leased address inside .113-.118, and do not lease the network address (.112), R-RO's gateway, the server's own address, or the broadcast (.119). Max users is at most 4.

Setting	Value
Server static IP	192.168.10.116
Subnet mask	255.255.255.248
Default gateway (pool setting)	192.168.10.122
Start IP address	192.168.10.113
Max users	4

Setting	Value
DNS Server (pool setting)	192.168.10.116

Then set Remote-PC1 and Remote-PC2's IP Configuration to DHCP.

Step 6: Add the Web-Eng server

Add a Server-PT on an Engineering access port (one of SW-B Fa0/1–Fa0/3). Give it the static address from Step 4, then turn on Services > HTTP so other departments have something to open when they test inter-VLAN routing.

Setting	Value
Server static IP	192.168.10.68
Subnet mask	255.255.255.224
Default gateway	192.168.10.65
HTTP service	On

Checkpoint A: inter-VLAN test

From a Sales PC, ping Web-Eng's IP, then open that same address in the PC's browser. Both should work. This test does not need a static route.

A new trunk can spend about 30 seconds in Spanning Tree listening and learning. If the first ping fails, wait and try again before you change the configuration.

Why does this work with no static route?

Because the two end devices are connected to the same router. If they were from different router, then you need to do a hop. In this case we don't need to.

Checkpoint B: MO-to-RO test

From a Sales PC, ping a Remote PC. That ping should succeed. The Lab 3 static routes already cover Sales and the Remote LAN. If it fails, those routes are missing from the routers. Enter every route in the Step 7 table, not only the blank row, then test again.

From a Finance PC, ping a Remote PC. That ping should fail. R-MO can forward the packet toward the Remote LAN, because Finance is connected on R-MO and the Lab 3 route to Remote is already there. R-RO has no route back to the Finance subnet, so the reply never returns.

What is missing, and on which router?

192.168.10.128/28 via 192.168.10.121 on R-RO

Step 7: Static route for Finance

The four routes already printed below are unchanged from Lab 3. Leave them in place. The only new route is the blank row, on R-RO, for the Finance network from Step 1. Use the same next hop R-RO already uses for the other MO subnets (192.168.10.121).

Router	Destination network	Mask	Next hop
R-MO	192.168.10.112	255.255.255.248	192.168.10.122
R-RO	192.168.10.0	255.255.255.192	192.168.10.121
R-RO	192.168.10.64	255.255.255.224	192.168.10.121
R-RO	192.168.10.96	255.255.255.240	192.168.10.121
R-RO	192.168.10.128	255.255.255.240	192.168.10.121

With the routes in place, test MO-to-RO again. Ping from a Sales PC to a Remote PC, and from a Finance PC to a Remote PC. The Sales ping can succeed while the Finance route is still missing, so the Finance ping is the one that checks the new route.

Step 8: Fundamental questions

1. Why can one physical link between R-MO and SW-A carry traffic for four separate subnets?

Because each subinterface handles a VLAN's tagged traffic. The switch sends frames tagged with the VLAN using trunking. The router then uses the VLAN ID on each subinterface to decide which Layer 3 network the traffic belongs to.

2. What is the difference between an access port and a trunk port?

Access ports are used by end devices such as pc and servers. They can only carry one kind of vlan. The trunk port is used by routers and switches. They can carry multiple vlans at once.

3. Why does Sales-to-Web-Eng traffic need no static route, while traffic between MO and the Remote office does?

Because Sales-to-Web-Eng doesn't need to pass through different routers. Static routing is used when you want to "hop" from one router to another.

4. Why does a DHCP pool typically include a DNS server address alongside the IP range it hands out?

Because it is from the same network. It also automatically configures the network for end devices. So you don't need to manually configure on static
